R15/R13

Code No: 123CT/113BS

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, March - 2021
DIGITAL LOGIC DESIGN
(R15 - Computer Science and Engineering; R13 - Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Solve for x:
i) $(367)_8=(x)_2$ ii) $(378.93)_{10} = (x)_8$ iii) $(B9F.AE)_{16} = (x)_8$ iv) $(16)_{10} = (100)_x$.
b) Convert $(163.875)_{10}$ to binary, octal, hexadecimal. [8+7]

2.a) What are universal gates? Realize AND, OR, NOT, XOR gates using universal gates.
b) Obtain the canonical SOP form of the following functions.
i) Y (A,B) = A+B ii) Y(A,B,C,D) = AB+ACD [7+8]

3.a) Simplify the expression $Y=\sum m(7,9,10,11,12,13,14,15)$ using the k-map method.
b) Simplify the following Boolean function using K-map:
$F(A,B,C,D)= \sum m(1,3,7,11,15) + \sum d(0,2,5)$ [8+7]

4.a) Simplify the expression $Y= \pi (7, 9, 10, 11, 12, 13, 14, 15)$ using the k-map method.
b) Simplify the expression $Y= m_1 + m_5 + m_{10} + m_{11} + m_{12} + m_{13} + m_{15}$ using the k-map method. [7+8]

5.a) Draw the schematic diagram and truth table for half adder. Explain the design approach for half adder using universal gates. Draw the logic diagrams with relevant expressions.
b) Design, draw and explain a 4-bit binary carry look ahead adder. [8+7]

6.a) Describe the operations performed by the following logic circuits with an example
i) Comparator ii) Decoder iii) Encoder
b) Explain the operation of a 3-to-8 decoder 74LS138. Realize 4-to-16 decoder using two 3-to-8 decoders. [8+7]

7.a) Design, draw and explain a synchronous MOD-12 down-counter using j-k flip-flop.
b) Design, draw and explain a 4-bit ring counter using D- flip flops with relevant timing diagrams. [7+8]

8.a) Give the logic implementation of a 32×4 bit ROM using decoder of suitable size.
b) Implement the following Boolean function with PLA $F(A,B,C) = \sum m(1,5,6,7)$. [7+8]

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